

Building an Intelligent Nation: Addressing the Core Challenges of Informatization

Executive Summary

Informatization—the transformation of a nation's economy, governance, and society through integrated information systems—is essential for competitiveness and innovation. However, progress is often hindered by structural, human, and policy barriers. This brief identifies five critical issues and outlines actionable strategies to achieve sustainable national digital transformation.

1. Data Fragmentation and Lack of Interoperability

Information silos across ministries and enterprises prevent effective data use. Without interoperability, systems remain fragmented, duplicative, and inefficient.

Recommendations:

- 1 Establish national data interoperability standards (APIs, metadata, identifiers).
- 2 Develop a National Data Exchange Platform for secure sharing.
- 3 Implement the 'data once' principle across all government systems.

2. Insufficient Human Capital and Digital Literacy

Many civil servants, workers, and citizens lack the digital skills to manage and interpret information effectively, leading to slow adoption and inefficiency.

Recommendations:

- 1 Launch a National Informatics Academy for government and industry training.
- 2 Integrate informatics and data literacy into educational curricula.
- 3 Promote public awareness programs on responsible data use.

3. Cybersecurity, Privacy, and Trust Deficits

Expanding connectivity increases vulnerability to cyberattacks and data breaches, risking national security and eroding public trust.

Recommendations:

- 1 Create a National Cybersecurity Operations Center (SOC).
- 2 Enforce privacy-by-design standards and conduct regular audits.
- 3 Introduce transparent data protection laws and citizen consent frameworks.

4. Weak Governance and Policy Coordination

Overlapping authorities and unclear data ownership cause duplication and inefficiency. Central coordination is critical to national progress.

Recommendations:

- 1 Establish a Central Informatics Authority to coordinate across ministries.
- 2 Enact a National Data Governance Act defining rights and oversight.
- 3 Create an inter-ministerial council for digital policy alignment.

5. Lack of Long-Term Strategy and Sustainability

Many informatization programs are pilot-based and lack strategic continuity, leading to resource waste and inconsistent progress.

Recommendations:

- 1 Develop a 10-year National Informatization Master Plan with clear KPIs.
- 2 Secure sustainable funding via public-private partnerships.
- 3 Measure progress through an annual Informatization Index.

Cross-Cutting Challenges (2025–2030)

- 1 AI governance and algorithmic transparency.
- 2 Data sovereignty and national cloud independence.
- 3 Green informatics and IT energy management.
- 4 Digital inclusion for rural and low-income populations.
- 5 Quantum readiness and post-quantum security.

Conclusion

Informatization success depends on technology, governance, skills, and public trust. Addressing these issues holistically will build a secure, data-driven, and inclusive digital society that supports innovation and sustainable growth.